DEPARTMENT 1 · DEVELOPMENT AND IT · PROFESSION 10

AI consultant

Profession 10 of 10 in Development and IT.

What the profession means

An AI consultant helps a business decide where AI will help, how much it will return and how to implement it safely. The consultant audits processes, proposes solutions, calculates the return, and supports implementation and training.

What professionals do every day

Interviews staff, maps processes, finds opportunities, estimates benefit and cost, writes proposals, runs a proof of concept, supervises implementation and trains the team.

Why businesses hire them, and what problems they solve

Many businesses are unsure where to start and afraid of wasting money. A consultant gives a clear, costed plan with a small first step.

- Time lost on repetitive work.
- Unclear priorities and expensive tool purchases.
- Risk: data, accuracy and staff concerns.

Skills you need

- Process mapping and opportunity scoring.
- ROI and payback calculations (below).
- Proposal writing and client communication.
- AI risks: accuracy, privacy, bias, change management.
- Training and handover.

Tools (free and paid)

Tool	What it is for	Free option	Paid option	Official link (not checked live)
Microsoft Excel	Models and ROI tables	Free online version	Paid plans	www.microsoft.com/microsoft-365/excel
Google Sheets	Shared models	Free	Paid Workspace plans	sheets.google.com
Miro	Process maps and workshops	Free tier	Paid plans	miro.com
n8n	Prototype automations	Self-hosted free	Cloud plans	n8n.io

Learning sequence: beginner, intermediate, advanced

Level	What you can do when you finish

Beginner	Estimate the saving for one process.
Intermediate	Write a one-page opportunity summary.
Advanced	Run a two-week proof of concept with clear success criteria.

Projects by level

- **Beginner:** Real estate: automate enquiry handling and site visit booking.
- **Intermediate:** Education: automate course enquiries and attendance reminders.
- **Advanced:** Retail: automate stock alerts and supplier orders.

Client-ready portfolio project

A complete AI automation proposal for a fictional real estate company: process map, opportunity list, ROI calculation, risks, a two-week pilot plan and a training outline.

Worked example

Worked example. Staff spend 5 hours a week copying enquiries. At Rs 400 an hour over 48 working weeks, the saving is Rs 96,000 a year. Subtract a tool cost of Rs 24,000 a year and the build cost gives the payback period and return. The calculator in Module H shows the arithmetic.

Assignment with full solution

ASSIGNMENT

Estimate the return on an automation

Business problem: A firm asks whether automating enquiry replies will pay for itself.

Steps

1. List the hours saved per week, the hourly cost, and the working weeks per year.
2. Enter the build cost and the monthly tool cost.
3. Compute the first-year net, the payback period and the return.
4. State the assumptions and what would change the answer.
5. Show the result as a range (low and high hours saved), not one number.

Expected output: A table of results with assumptions, a payback period and a range.

SOLUTION (ANSWER KEY)

Every figure is an assumption until measured. The calculator returns the same numbers as the worked example: a saving of Rs 96,000, a first-year net of Rs 12,000, payback in 10 months, and a return of 14.3%. Present a range, because the hours saved are an estimate.

PYTHON

```
def automation_roi(hours_saved_per_week, hourly_cost, weeks_per_year, build_cost,
monthly_tool_cost):
    annual_saving = hours_saved_per_week * hourly_cost * weeks_per_year
    annual_tool = monthly_tool_cost * 12
    first_year_net = annual_saving - build_cost - annual_tool
    net_per_year_after_build = annual_saving - annual_tool
```

```
    payback_months = build_cost / (net_per_year_after_build / 12) if net_per_year_after_build
> 0 else float("inf")
    roi_percent = first_year_net / (build_cost + annual_tool) * 100
    return {"annual_saving": annual_saving, "annual_tool": annual_tool,
            "first_year_net": first_year_net, "payback_months": round(payback_months, 1),
            "roi_percent": round(roi_percent, 1)}

if __name__ == "__main__":
    r = automation_roi(hours_saved_per_week=5, hourly_cost=400, weeks_per_year=48,
                       build_cost=60000, monthly_tool_cost=2000)
    assert r["annual_saving"] == 96000 and r["annual_tool"] == 24000
    assert r["first_year_net"] == 12000
    assert r["payback_months"] == 10.0
    assert r["roi_percent"] == 14.3
    print("roi tests passed:", r)
```

Common mistakes

- Promising a return without measuring the current process.
- Ignoring the cost of staff time during implementation.
- No pilot, so the full project starts before anyone has tested it.
- Ignoring privacy and data-handling rules.

Freelancing opportunities

- AI audits and automation roadmaps for small and mid-size firms.
- Pilots, implementation oversight and staff training.
- Freelance briefs for 'AI consultant', 'AI strategy', 'process automation', 'ROI analysis'.

Learning resources

LEARNING RESOURCES (VERIFY BEFORE USE)

- AI for Everyone (Coursera, check current availability and fees)
- NIST AI Risk Management Framework

Part of **Master Freelancing: AI and Automation**. Back to the chapter list on skillnavora.in · Open the full textbook. Official links in this chapter were not checked live; confirm before relying on them.